STM32H745ZI — Pin Assignment

PINOUT REV 2026-07-03

Triple Inverted Pendulum on Cart · Nucleo-H745ZI-Q dual-core · ODrive S1 / Ethernet / ESP32

M7 — Cortex-M7, bare-metal control
M4 — Cortex-M4, FreeRTOS
Planned / not yet wired

Encoders — SPI1 (M7)

PA5	SPI1_SCK	AS5048A clock · 6.25 MHz, mode 1
PA6	SPI1_MISO	AS5048A data in
PD7	SPI1_MOSI	AS5048A command out (0xFFFF) **— verify vs PB5**
PD14	ENC_CS0	Encoder A chip-select (CS1/CS2 alias here)

UART Links (M7 / M4)

PD5	USART2_TX	M4 → ESP32-S3 display · 921600, DMA
PD6	USART2_RX	M4 ← ESP32-S3 display
PD8	USART3_TX	M7 · ST-LINK virtual COM port
PD9	USART3_RX	M7 · ST-LINK virtual COM port

Ethernet — RMII + LWIP (M4)

PA1	ETH_REF_CLK	LAN8742 PHY (onboard)
PA2	ETH_MDIO	mgmt data
PA7	ETH_CRS_DV	carrier / data valid
PC1	ETH_MDC	mgmt clock
PC4	ETH_RXD0	rx data 0
PC5	ETH_RXD1	rx data 1
PB13	ETH_TXD1	tx data 1
PG11	ETH_TX_EN	tx enable
PG13	ETH_TXD0	tx data 0 · STM32 @ 192.168.1.10

No-pin Peripherals

TIM7	M7	5 kHz bare-metal control loop
IWDG2	M4	independent watchdog (~2 s)
WWDG2	M4	window watchdog (~20 ms)
TIM6	M4	FreeRTOS HAL timebase

ODrive S1 — FDCAN1 (M7)

PD0	FDCAN1_RX	from ODrive · 1 Mbps classic CAN
PD1	FDCAN1_TX	to ODrive

Operator I/O (M7)

PA3	ADC1_INP15	Potentiometer wiper
PC13	B1 (GPIO_IN)	Blue user button (active HIGH)
PB0	LD1 (GPIO_OUT)	Green LED · active control mode
PE1	LD2 (GPIO_OUT)	Yellow LED · position ctrl / tick
PB14	LD3 (GPIO_OUT)	Red LED · needs-index / fault
PE2	GPIO_OUT	Scope debug · 5 kHz loop pulse **(code-only)**

Clocks / System

PH0/PH1	RCC_OSC	HSE main oscillator
PC14/15	RCC_OSC32	LSE 32.768 kHz

USB OTG FS (M7 · unused)

PA11/12	USB_DM / DP	data lines
PA8/PA9	SOF / VBUS	frame / bus sense
PD10	PWR_EN	port power enable (out)
PG7	OVCR / GPXTI7	over-current (EXTI)

Rail End-Stops (M7 · planned)

PG2	EXTI2	Home switch · NC, pull-up, rising-edge
PG3	EXTI3	Far switch · NC, pull-up, rising-edge
		enable at RAIL_HOME_SWITCH_WIRED=1

Notes & flags

1. **SPI1_MOSI = PD7** in the current .ioc — differs from the earlier "PB5" decision and HANDOFF (PB5 / Zio D11). The working encoder implies the flashed MOSI matches the physical wire; confirm a rebuild-from-.ioc keeps it on the wired pin.
2. **Encoders B & C** have no dedicated CS pins yet — the driver aliases CS1/CS2 to PD14 until they are wired.
3. **PE2** (scope debug) is configured in main.c user code, not the .ioc — it will not appear in CubeMX.
4. **PG2 / PG3** rail switches are planned only: not in the .ioc, EXTI compiled out until wired and RAIL_HOME_SWITCH_WIRED=1.